

Hekmat Kawas

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EDUCATION

Bachelor of Engineering in Chemical Engineering

Toronto Metropolitan University

Expected Dec 2026

Toronto, ON

Coursework: Separation Processes, Process Modeling & Simulation, Equipment Design, Particulate Engineering, AI in Chem Eng

EXPERIENCE

Junior Metallurgist - Process Research Ortech Inc.

May 2026 – Present

- Optimized Mg extraction in a pilot-scale HCl leach circuit by testing acid concentrations from 10–20%, identifying **16% as optimal** and raising extraction from **89% to 91%**
- Redesigned pilot filter press setup for leach residue separation, expanding filtration area from 8 to 12 plates and revising the operating procedure to **cut filtration time by 23%**
- Co-designed and scaled extraction processes for magnesium chloride recovery, introducing a water-wash and vacuum filtration protocol that **recovered 93% purity silica** from the process residue
- Diagnosed and repaired leaking lines and miscalibrated dosing pumps across pilot plant and lab unit operations, cutting chemical consumption by **5%** and reducing downtime by **1 hour/day**
- Wrote **3 SOPs** for pilot plant HCl charging and acid transfers and trained a new hire on safe acid handling
- Co-authored a **client technical report** on pilot-scale MgCl₂ leach results, compiling test data and mass balances

R&D Lab Technician - Protech Group

Jan 2025 – Aug 2025

- Prepared and validated powder coating formulations using mixing, extrusion, and grinding, including **75+** that advanced to full-scale production, generating 70 tons annually and **\$725K+ in recurring revenue**
- Resolved 5+ contamination cases by systematically isolating raw materials, equipment, and process steps, raising batch pass rates by **20%** and freeing lab capacity that added **\$15K+ in profit**
- Analyzed Excel batch data on resin ratio, cure time, and pigment load to find patterns in successful trials, cutting trial iterations and saving **5 hours per formulation**
- Digitized manual batch records and lab documentation for **200+ formulations**, cutting record retrieval time and reducing downtime by 7% (**\$10.5K+ annual savings**)
- Diagnosed dispersion and formulation-balance failures in client samples, adjusting formulations, grinding and baking times, and extruder settings until samples passed client evaluation

PROJECTS

Lithium Recycling Plant Design (Capstone)

- Sized all equipment and piping**, including 2 bioleaching batch reactors (54 m³ each), 4 membrane filter presses, and 35 pipelines and pumps with pressure drop and NPSH checks
- Performed mass balances for a 15,000 t/yr battery bioleaching plant, achieving **~86% Li recovery** as Li₂CO₃ (~1,930 t/yr)
- Completed energy balances for **15 major units**, including 443 kW aeration compressors and a 325 kW heat exchanger

Process Equipment Sizing Program

- Developed a Python-based process equipment sizing application integrating fluid mechanics and reaction engineering to evaluate pump performance, pipe hydraulics, and reactor sizing under user-defined conditions
- Implemented core chemical engineering design correlations, including flow regime analysis, friction factor correlations, and total dynamic head calculations within an interactive user interface
- Generated equipment sizing and performance estimates to support preliminary design and feasibility studies

Liquid-Liquid Blending Unit Design

- Engineered a stirred-tank blending system for miscible liquids, optimized through impeller design and tank geometry
- Modeled equipment in SolidWorks and evaluated mixing geometry to support uniform concentration at steady state
- Evaluated and selected process control strategies for optimal concentration consistency and minimal composition variability

TECHNICAL SKILLS

- Process Engineering:** Mass & energy balances, equipment & piping sizing, PFDs, process optimization, pilot-scale operations
- Software & Programming Languages:** Aspen Plus, Aspen HYSYS, Python, SolidWorks, Excel, MATLAB, Java
- Lab & Process Skills:** Leaching, filter press & vacuum filtration, hazardous material handling (WHMIS), SOP development